

Yikun He

Nov 2004 | WeChat:fisher001122 | heyikun98@gmail.com
github.com/heyikun98-design

EDUCATION

Central South University | Dundee International Institute

Sep 2023 – Present

BEng in progress | Mechanical Engineering with Transportation

Joint programme with the University of Dundee; fully English-taught curriculum

English: IELTS 7.0 overall (Listening 8.5).

Relevant coursework: Control Systems, Structural Dynamics, Mechanics of Materials, Finite Element Analysis, Linear Algebra, Probability and Statistics.

University of California, Berkeley | Summer Sessions

Summer 2024

Visiting student | Creative Nonfiction (A); ESL Berkeley Experience (Pass)

INTERNSHIP EXPERIENCE

Mimicverse | Robotics R&D Intern

17 Aug 2026 – Present

Mechanical Design, Simulation and Reinforcement Learning | Shenzhen, China

- **Policy training & simulation:** Reproduced an open-source reinforcement learning workflow for robot policy training and deployment; exported URDF models and conducted simulation analysis.
- **Mechanical design:** Designed leg mechanisms with reference to Microduck, added wrist yaw, and enlarged the torso to accommodate PCBs as part of small humanoid robot design iterations.
- **Prototyping:** Built an open-source Microban robot through 3D printing, assembly and troubleshooting; refined the structure based on assembly feedback and manufacturing requirements.

Shenzhen Institutes of Advanced Technology, CAS (SIAT)

Summer 2026

Exoskeleton Research Intern | Gait Data Acquisition and Preprocessing

- Collected synchronized gait kinematics and surface EMG data using Vicon Nexus and Delsys; organized and preprocessed experimental data for exoskeleton research.

RESEARCH & PROJECT EXPERIENCE

PT-HICnet | Deep Learning for Collision Injury Prediction

Sep 2025 – Present

Individual Research Project | Supervisor: Assoc. Prof. Fan Li, Central South University

- **Model design:** Proposed replacing the PointNet++ backbone with Point Transformer to predict Head Injury Criterion (HIC) from 3D vehicle point clouds.
- **Implementation:** Used PyTorch to modify, train and evaluate the model; compared alternative backbones and investigated data and training issues.
- **Interim results:** Improved on the original baseline on a small dataset; drafted the methods section, with validation on an expanded dataset ongoing.

External Structural “Airbag” for Vehicle Collision Protection

2025

Team Member | Concept Development

- Co-developed an extendable front-end protection concept to increase energy-absorption stroke and mitigate collision impact; awarded Third Prize in a university-level transportation technology competition.

Structural Dynamics and Vibrations | Training Programme

Feb – Aug 2025

Participant | Course Instructor: Prof. Jiann-Wen Woody Ju, UCLA

- Completed 54 hours of English-taught coursework (96.60/100), performed structural and transient-response analyses in ANSYS, and attended in-person research training at Shanghai Jiao Tong University.

RESEARCH PAPER

Fan Li, Shuang Li, Ziyuan Zhou, **Yikun He**, Ke Li. “Batch Multi-fidelity Bayesian Optimization via Nonlinear Autoregressive Gaussian Process.” IEEE Transactions on Evolutionary Computation. Conditionally accepted; fourth author.

TECHNICAL SKILLS

Algorithms: Point-cloud deep learning model design, training and evaluation; reinforcement learning fundamentals and open-source training experience.

Programming: Python, PyTorch, Linux, Git.

Robotics tools: SolidWorks, AutoCAD, ANSYS, MuJoCo, mlab; URDF modeling, 3D printing and assembly; basic ROS 2.

Mathematics: Linear algebra, probability and statistics, differential equations and numerical methods.